Complex-Rotation Manifold Routing

Continuous Phase Control for DDoS Isolation in Name-Based Networks

Milan Nikolic
Independent Researcher
m_nikolic@yahoo.com

Abstract

Name-based forwarding collapses resolution into forwarding, but its metric space is usually static: a name is matched to a prefix, and a prefix to an instance. When an attacker shares the legitimate prefix, prefix-only routing cannot separate the two. Discrete dynamic-dimensionality routing (DDMR) answers this by jumping from a one-dimensional prefix metric to a four-dimensional metric that includes latency, jitter and a local anomaly score; in earlier work it preserved legitimate delivery under a sustained DDoS, at the price of a hard state switch with hysteresis. We ask whether the switch can be replaced by a continuous rotation. Complex-Rotation Manifold Routing (CRMR) carries a phasor $\psi(t) = R(t)e^{i\theta(t)}$ whose phase rotates the metric between the prefix axis and the defensive axes, following a damped second-order law driven only by the local anomaly, and migrates legitimate flows to a replica with probability $\sin^2\theta$. We evaluate CRMR against static T-NDN and discrete DDMR on a 100-node, 10,000-name fabric over 30 paired seeds, under a sustained and a flapping DDoS. Under the sustained attack CRMR holds the legitimate content delivery rate (CDR) at 97.24% against 99.68% for DDMR and 73.82% for static T-NDN, while reducing the peak per-tick reconfiguration share from 1.00 to 0.157 (an 84% reduction) and passing the energy clause (3.31 J of four-dimensional homology against 14.08 J of prevented-loss energy). Under a flapping attack the rotation cannot complete inside a 3 s burst, and the CDR cost rises to 11.2 points; a burst-duration sweep locates the boundary, with the worst case at twice the controller's settling time and the gap falling to 5.6 points at 15 s. Persistent homology separates the attack as a new H_0 component; no persistent H_1 loop is observed, so the isolation is zero-dimensional rather than a topological hole. Continuity buys reconfiguration smoothness, not free delivery: the paper quantifies the exchange rate.

Keywords: named data networking; DDoS isolation; persistent homology; dynamic dimensionality; complex phase control; service migration.

1 Introduction

Information-Centric Networking (ICN) and Named Data Networking (NDN) make the name the forwarding key: an Interest carries a name, and a router forwards it toward a producer or a cache [1]. Name-based routing protocols such as NLSR flood name prefixes and install one FIB entry per prefix [2]; fisheye-style scoping reduces control overhead by refreshing distant state less often [3]. In a service fabric the prefix is also the handle of a role-fluid service instance, so a moving or replicated service can be reached without a DNS round trip.

The weakness is equally structural. If an attacker sends to the same prefix as legitimate clients, any router that selects an egress by prefix alone sends both to the same instance. The attacker then consumes the instance's queue and drops legitimate requests, and no amount of prefix aggregation separates the two. The defence must use a signal that is not in the prefix.

Dynamic Dimensionality Manifold Routing (DDMR) does that by treating the routing space itself as a function of telemetry: in calm conditions the metric is the one-dimensional prefix metric; when a local rate anomaly crosses a threshold the space expands to four dimensions (prefix, latency, jitter, anomaly), the attack forms its own component, and legitimate traffic is sent to a healthy replica. The transition, however, is a switch, with hysteresis and a detection window. A switch is a discontinuity in the control state: every flow changes egress in the same tick, the fabric must absorb a reconfiguration burst, and the state can chatter around the threshold.

This paper asks whether the switch can be replaced by a rotation. We propose Complex-Rotation Manifold Routing (CRMR): the fabric carries a phasor $\psi(t) = R(t)e^{i\theta(t)}$, whose phase $\theta \in [0, \pi/2]$ rotates the metric continuously from the prefix axis to the defensive axes, and whose amplitude R weights the defensive coordinates. The phase follows a damped second-order law driven only by the local anomaly. Routing is phase-aligned: the legitimate share sent to a healthy replica is $\sin^2\theta$, so migration is spread over the rotation instead of happening in one tick.

The paper makes five contributions.

1. A continuous formulation of dynamic-dimensionality routing: a phasor state, a damped rotation law, a one-parameter family of metrics, and phase-aligned probabilistic egress selection. We show explicitly that a rigid rotation of the point cloud would be an isometry and a no-op, so the rotation must act on the metric weights.
2. A paired-seed evaluation against static T-NDN and discrete DDMR on a 100-node, 10,000-name fabric with two attack regimes (sustained and flapping), 30 seeds, identical traffic, instances and capacity.
3. A quantified boundary: the CDR cost of continuity is negligible for long bursts and peaks when the burst duration matches the rotation's settling time. A burst-duration sweep from 1 to 15 s measures the exchange rate.
4. A negative topological result: the phase-weighted complex separates the attack as a new H_0 component but produces no persistent H_1 loop, so the isolation is zero-dimensional; the earlier "hole" language is not supported by the data.
5. An energy accounting with an explicit break-even: 3.31 J of four-dimensional homology against 14.08 J of prevented-loss energy under the sustained regime, which passes the clause only above 4.7 mJ per prevented request.

The result is deliberately two-sided. Under a sustained attack CRMR preserves legitimate delivery within 2.4 points of DDMR while removing 84% of the peak reconfiguration, at an unchanged latency. Under a 3 s flapping attack it pays 11.2 points of CDR for that smoothness. The paper reports both and explains why: a rotation cannot be faster than its own settling time, and a burst shorter than that settling time is not long enough to amortise the ramp.

2 Related work

Name-based forwarding. NDN forwards Interests by name and returns signed Data [1]. NLSR is the reference link-state protocol for name prefixes [2]; Fisheye State Routing refreshes near state often and far state rarely [3]. The fisheye overlay of Milić and Braun builds a peer view from measured RTT distances without an embedding [4]. The T-NDN plan used here is a static name-space construction with a fixed prefix metric and a persistent H_0 partition; its discrete dynamic extension is DDMR, which this paper compares against.

Topological data analysis. Persistent homology summarises the multi-scale connectivity of a point cloud [5]. Vietoris–Rips persistence is computed here by the `ripser` library. A filtration is a one-parameter family of simplicial complexes; CRMR makes that parameter a physical control variable, the phase θ, rather than a fixed radius.

What is new. To our knowledge the specific combination—a phasor-driven metric family, a damped rotation law in the control plane, and a probabilistic phase-aligned egress rule—has not been evaluated for DDoS isolation in a name-based fabric. The closest discrete mechanism is DDMR; the closest continuous mechanism is a weighted metric, which is what CRMR reduces to at a fixed θ. The contribution is the continuity and its measured price, not a new persistence algorithm: every phase is an ordinary real filtration, so the implementation reuses the M11 pipeline.

3 System model

3.1 State and telemetry

Each flow f observed by a router is described by the four-dimensional vector

$$S_f = \big[\, D_{\text{prefix}},\ L,\ J,\ A \,\big], \tag{1}$$

where D_{prefix} is the flow's position in the prefix hierarchy, L is the latency deviation from the flow's baseline path, J is an exponential moving average of inter-arrival jitter, and A is the local rate anomaly,

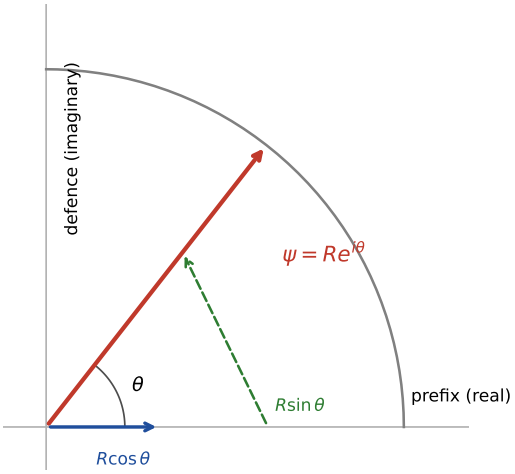


Figure 1: The CRMR phasor. The real axis is the prefix metric; the imaginary axis is the defensive metric. The phase θ rotates the fabric between them and the projections $\cos\theta$ and $R\sin\theta$ are exactly the weights in (4). A rigid rotation of the points would be an isometry; the rotation must act on these weights.

$A = \mathrm{clip}\big((\lambda - 2)/2, 0, 1\big)$ for a short-window rate λ in requests per second. The anomaly is computed per flow; the dimension is driven by a block-level monitor that freezes a nominal rate during a warm-up and maps the Poisson z-score of the current rate by $(z - 3)/3$ to $[0, 1]$.

3.2 The phasor

The fabric state is a complex number

$$\psi(t) = R(t)\, e^{i\theta(t)}, \qquad \theta \in [0, \pi/2], \quad R \geq 1, \tag{2}$$

whose phase θ blends the prefix axis ($\theta = 0$, the real axis) with the defensive axes ($\theta = \pi/2$, the imaginary axis), and whose amplitude R scales the defensive coordinates. The phase follows a damped second-order law

$$\ddot{\theta} + 2\zeta\omega\,\dot{\theta} + \omega^2\big(\theta - \theta^*(t)\big) = 0, \qquad \theta^*(t) = \frac{\pi}{2}\,\sigma\left(\frac{A(t) - A_0}{s}\right), \tag{3}$$

with $\zeta = 1$, $\omega = 3\,\mathrm{rad/s}$, and a saturated target. There is no attack flag and no clock in (3): the target is a function of the same anomaly the discrete controller uses. The amplitude follows $R(t) = 1 + \rho\,\mathrm{clip}(A, 0, 1)$ with $\rho = 1$. Figure 1 shows the construction: the phase is the angle between the prefix axis and the defensive axis, and the projections of ψ are the metric weights.

3.3 Metric family

For a fixed phase, the distance between two flows is

$$d_\theta^2(a, b) = \cos^2\theta\, d_{\mathrm{prefix}}^2(a, b) + \sin^2\theta\,\left(w_L d_L^2 + w_J d_J^2 + w_A d_A^2\right), \tag{4}$$

with equal weights $w_L = w_J = w_A = 1$. The complex number is a compact encoding of (R, θ), not a complex-valued homology: at every θ, (4) is a real weighted Euclidean metric and the Vietoris–Rips filtration is standard. In implementation the re-weighting is applied to the coordinates,

$$x'_j = \big[\cos\theta\, x_{j,0},\ R\sin\theta\, x_{j,1},\ R\sin\theta\, x_{j,2},\ R\sin\theta\, x_{j,3}\big], \tag{5}$$

and the filtration radius scales with the phase,

$$\varepsilon(\theta) = \varepsilon_0\sqrt{\cos^2\theta + 3R^2\sin^2\theta}, \qquad \varepsilon_0 = 0.25. \tag{6}$$

A remark that matters for implementation: a rigid rotation of the point cloud, $x_j \mapsto e^{i\theta}x_j$, would preserve all pairwise distances and leave every persistence diagram unchanged. The rotation must act on the metric weights, as in (5), or it is a no-op. The phase is therefore a genuine parameter of the topology, not a relabelling.

Table 1: Frozen scenario and controller constants.

Quantity	Value
Topology / names	100 nodes, 20 blocks × 500 names
Instances / capacity	3 per block, 200 requests/s, $20\,\text{ms} \cdot u^3$
Legitimate load	200 requests/s, block 0 share 0.40
Attack load	500 requests/s, 20 bots, block 0
Regimes	sustained $[30, 45]$ s; flapping 5×3 s
Control tick / anomaly window	0.1 s / 1 s (10 s warm-up)
Phase dynamics	$\zeta = 1$, $\omega = 3$ rad/s, $\rho = 1$
Phase thresholds	$A_0 = 0.2$, $A_{\text{hi}} = 0.8$, calm $\theta < 0.05$
Manifold	150 flows, $\varepsilon_0 = 0.25$, period 2 s
Seeds	30 paired (2030–2059); pilot 2026 excluded

3.4 Phase-aligned routing

At each control tick the router computes the H_0 components of the phase-weighted complex at $\varepsilon(\theta)$ and their centroids. A component whose mean anomaly and jitter exceed the defence thresholds is the attack component; flows whose nearest centroid is the attack component stay on the primary instance, so the attack cannot displace legitimate traffic onto the replica. Every other flow of an attacked block is sent to a healthy replica with probability

$$p(t) = \sin^2 \theta(t), \tag{7}$$

using a deterministic per-flow draw, so a flow keeps its choice while the phase is stationary and migrates smoothly as the phase rotates. The theoretical reconfiguration share $p(t)$ is the quantity we use to measure control churn.

3.5 Baselines

Static T-NDN runs the same anomaly monitor and the same manifold code with the dimension pinned to one, and always uses the primary instance. **DDMR** uses the discrete M11 controller: it jumps from one to four dimensions after two consecutive anomalous ticks, collapses after four calm ticks, and routes all legitimate traffic of an attacked block to the replica when the dimension is four. All arms share the network, instances, request stream and drop-tail capacity model; only the selection rule differs.

4 Methodology

Scenario. The network is the 100-node geometric fabric of the T-NDN study; names are grouped into 20 prefix blocks of 500 names. Each block has three instances (the primary and two replicas) with 200 requests/s capacity and a $20\,\text{ms} \cdot u^3$ queue. Legitimate Poisson traffic runs at 200 requests/s for 60 s, with block 0 carrying 40% of it. A botnet of 20 consumers sends 500 requests/s to block 0, so the attack and legitimate traffic share the base prefix. Two attack regimes are evaluated: *sustained* (one 15 s burst, $t \in [30, 45]$ s) and *flapping* (five 3 s bursts separated by 3 s gaps, $t \in [30, 57]$ s). The control tick is 0.1 s and the anomaly window is 1 s.

Pre-registration. Thresholds were frozen before the evaluation: peak reconfiguration at most 50% of DDMR's; sustained CDR gap at most 5 points and gain over static at least 15 points; energy below the prevented-loss energy. A pilot on seed 2026 fixed the values and is excluded; the evaluation uses seeds 2030–2059 (30 paired seeds). The flapping CDR gap is a measured boundary, not a pass criterion. The frozen protocol is in `docs/crmr-predictions.md` of the accompanying repository.

Metrics. Legitimate CDR is the delivered fraction of legitimate requests in the attack windows, overall and for the attacked prefix (block 0); p95 latency is computed over delivered legitimate requests; peak reconfiguration is the largest single-tick change of $p(t)$; energy is the four-dimensional homology time at 15 W against 0.02 J per prevented legitimate loss.

Table 1 lists the frozen constants.

Table 2: 30-seed means under the two attack regimes (legitimate traffic). CDR is the fraction of legitimate requests delivered in the attack windows; block 0 is the attacked prefix; peak churn is the largest single-tick change of the reconfigured legitimate share.

Regime	Arm	CDR (%)	CDR_0 (%)	p95 (ms)	Peak churn
sustained	Static T-NDN	73.82 ± 0.34	34.39 ± 0.55	132.0	0.00
sustained	DDMR	99.68 ± 0.07	99.19 ± 0.17	124.0	1.00
sustained	CRMR	97.24 ± 0.31	93.08 ± 0.79	124.7	0.157
flapping	Static T-NDN	73.93 ± 0.30	34.75 ± 0.59	132.3	0.00
flapping	DDMR	97.98 ± 0.44	94.93 ± 1.11	125.5	1.00
flapping	CRMR	86.79 ± 0.48	66.92 ± 1.15	129.3	0.157

5 Results

5.1 Sustained attack: continuity at a small price

Table 2 reports the 30-seed means. Under the sustained attack CRMR holds the legitimate CDR at $97.24\% \pm 0.31$ against $99.68\% \pm 0.07$ for DDMR and $73.82\% \pm 0.34$ for static T-NDN. The gap to DDMR is 2.44 points (95% CI ± 0.30), inside the 5-point guardrail on all 30 seeds, and the gain over static is 23.42 points (minimum seed 19.59). On the attacked prefix CRMR delivers 93.08% against 99.19% and 34.39%; on the whole legitimate stream it delivers 99.31% against 99.92% and 93.45%. The p95 latency is unchanged (124.7 ms versus 124.0 ms for DDMR and 132.0 ms for static), because the rotation moves traffic to a replica rather than onto a slower path.

5.2 Flapping attack: the rotation's bill

Under 3 s flapping bursts the picture changes. CRMR delivers 86.79% against 97.98% for DDMR and 73.93% for static, a gap of 11.19 points (seed range 7.06–13.40). Figure 2 shows the phase and the resulting legitimate share for the base evaluation seed. The per-burst decomposition in Fig. 3 shows why: each burst starts with the phase near calm, and the rotation needs roughly one settling time to move the legitimate share from zero to one. The loss is concentrated in the first second of every burst. The control churn is nonetheless exactly what the design promised: the peak single-tick reconfiguration share is 0.157 against 1.00 for DDMR, an 84% reduction, on every seed. The total variation of the share is, however, the same for both arms (10.02 versus 10.07 per run): continuity redistributes churn over time, it does not remove it. This is the central empirical point of the paper, and it is not the one we expected when we started.

5.3 Where continuity stops paying: the burst-duration sweep

To locate the boundary we repeated the experiment with a single burst of length $T \in \{1, 2, 3, 4, 5, 7, 10, 15\}$ s (10 seeds per point). The attacked-prefix CDR is in Fig. 4. The CRMR gap to DDMR is 28.6 points at $T = 1$ s, peaks at 39.6 points at $T = 2$ s, and falls monotonically to 5.6 points at $T = 15$ s. The maximum at $T \approx 2$ s is the signature of the controller's settling time: a burst shorter than the rotation ends before the ramp matters, and a burst much longer amortises it. The worst case is therefore not the shortest attack but the one that matches the control loop. This gives a design rule: choose ω so that $4/(\zeta\omega)$ is well below the shortest attack duration that must be survived, or accept the measured gap.

5.4 The topological signature

Figure 5 shows the Betti numbers of the phase-weighted complex for a sustained run. After the first manifold epoch, calm conditions give $\beta_0 = 1$; when the attack starts the complex splits into two and then three components, and β_0 returns to one after the attack. The attack is isolated as a *component*, not as a loop: across ten evaluation seeds and 7,289 phase-active epochs, β_1 is zero in every CRMR epoch (the discrete DDMR complex produces a short-lived H_1 feature in 2.1% of flapping and 6.3% of sustained shock epochs, with no persistence beyond one epoch). The "topological hole" language of the motivating brief is therefore not what the data show; the robust invariant is H_0, and the router uses exactly that. This is a negative result for the hole hypothesis, reported as such.

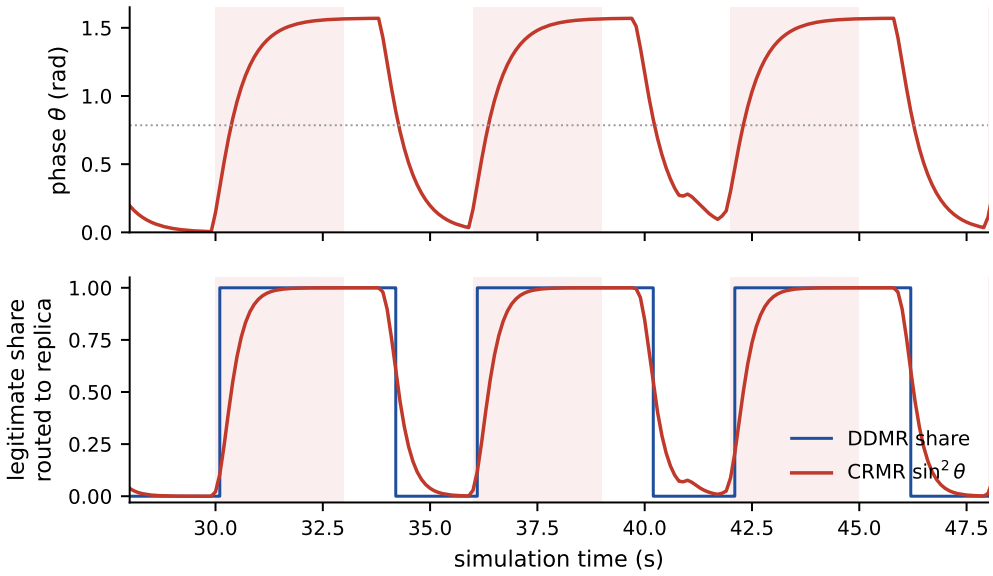


Figure 2: Flapping regime (evaluation seed 2030). Top: phase $\theta(t)$ rotates from calm to defence and back inside each burst; the dotted line is the gate used for the crossing count. Bottom: the legitimate share sent to a replica. DDMR switches in one tick; CRMR follows $\sin^2\theta$, spreading the migration over roughly one second. Shaded bands mark attack bursts.

Table 3: Pre-registered H15 criteria on the 30 evaluation seeds. The flapping CDR gap was declared a measured boundary in advance, not a pass criterion.

Criterion	Value	Limit	Verdict
Peak reconfiguration vs DDMR	0.157	≤ 0.50	pass
Sustained CDR gap vs DDMR	2.44	≤ 5 points	pass
Sustained CDR gain vs static	23.42	≥ 15 points	pass
Flapping CDR gap vs DDMR	11.19	measured	reported
Energy, sustained (CPU/saved)	0.235	< 1.0	pass

5.5 Churn and energy

The peak reconfiguration is the headline mechanism result: 0.157 versus 1.00, a factor 6.4 and an 84% reduction, with the same number of security-relevant state transitions (10.13 per flapping run for both arms; 2.27 versus 2.40 for sustained). What the rotation removes is the thundering herd at each edge, not the need to reconfigure.

The four-dimensional homology cost is 3.31 J per sustained run against 14.08 J of prevented-loss energy (ratio 0.235, 30/30 seeds pass); under flapping it is 4.44 J against 7.70 J (ratio 0.577, 30/30 pass). The break-even is 4.7 mJ per prevented request under the sustained regime and 11.5 mJ under flapping. At the stipulated 0.02 J per loss the clause passes with an order of magnitude of margin; if a prevented loss is worth less than 4.7 mJ the energy clause would refute the mechanism, exactly as the anti-cheat clause of the original brief demands to be checked. Table 3 collects the pre-registered criteria.

6 Discussion

The exchange rate of continuity. The experiments give a clean trade. For a sustained attack, rotating the metric costs 2.44 points of legitimate CDR and buys an 84% reduction in peak reconfiguration, at the same latency and a modest energy cost. For a 3 s flapping attack, the same rotation costs 11.19 points, because the ramp is a larger fraction of the burst. The burst sweep turns this into a quantitative design rule: the gap peaks when the burst duration is comparable to the controller's settling time and decays as the burst lengthens. A network that expects sustained DDoS pressure should rotate; a network that expects short bursts should switch, or rotate faster and accept the higher per-tick churn.

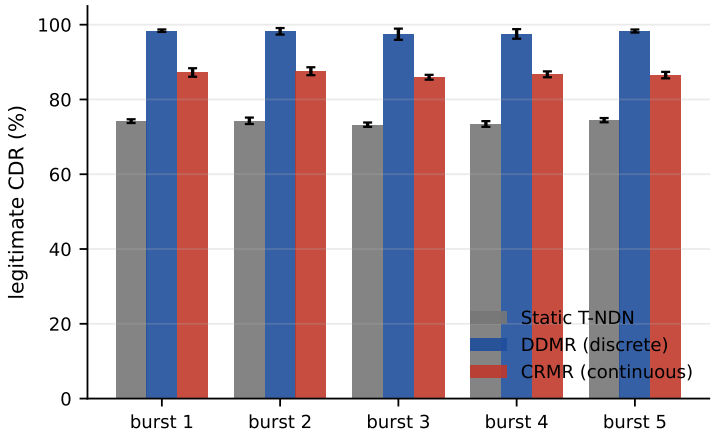


Figure 3: Per-burst legitimate CDR under the flapping regime (mean and 95% CI over 30 seeds). The CRMR loss is confined to the ramp at the start of each burst; the static baseline is flat at the capacity-limited floor.

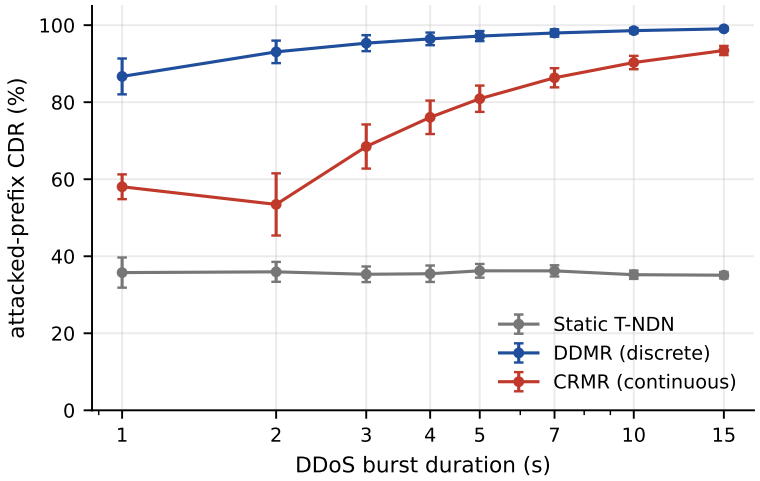


Figure 4: Attacked-prefix CDR against burst duration (single burst, 10 seeds per point, mean and 95% CI). Static T-NDN is capacity-limited; DDMR jumps to the replica after detection; CRMR pays the rotation ramp, with the worst gap at about twice the settling time and the gap vanishing as the burst lengthens.

Why the peak matters. The total reconfiguration is invariant (10.02 versus 10.07), so the benefit is not fewer changes but smaller ones. A router that must reprogram a forwarding table, signal a replica, or drain a queue cares about the size of a single reconfiguration, not only its integral. Spreading a 100% switch over one second reduces the instantaneous load on the control plane and on the replica's admission control. This is the operational meaning of the 6.4× peak reduction, and it is testable in a packet-level implementation; ours is not.

What the topology does and does not do. The phase-weighted complex splits the attack into its own H_0 component and the router uses the component boundary to decide which flows may move. It does not produce persistent H_1 loops, and no part of the mechanism depends on them. The value of persistent homology here is a robust, parameter-light partition (components and centroids), not a hole detector. Papers that claim a "topological hole" should measure β_1; ours does not support the claim.

Limitations. The evidence is a control-plane and capacity model, not a packet trace. The attack signature is stipulated: bots send 25 requests/s each with high jitter, which the four-dimensional space separates by construction. A low-rate, jitter-mimicking attack would not separate on these axes and was not tested. The capacity model is a per-instance drop-tail queue; there is no PIT, no cache, no per-packet strategy and no link-level congestion. The fabric assumes a healthy replica with spare capacity; if all

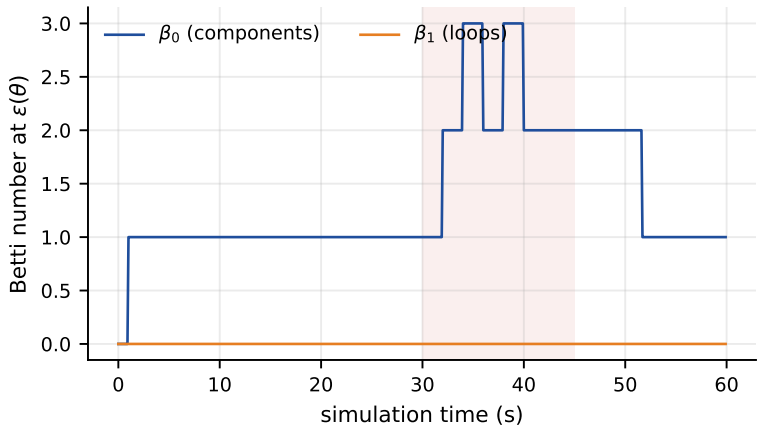


Figure 5: Betti numbers of the phase-weighted complex under the sustained regime (seed 2030). H_0 splits during the attack; H_1 is zero throughout. The isolation is zero-dimensional.

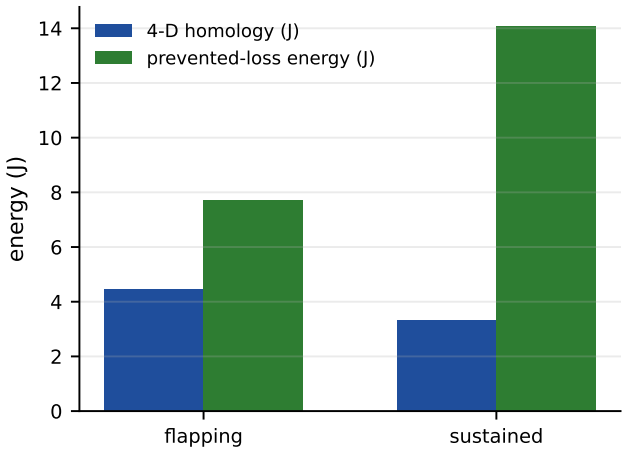


Figure 6: Energy accounting. The four-dimensional homology cost (blue) against the energy of the prevented legitimate losses (green) under the two regimes.

three instances are attacked the mechanism has nowhere to move traffic. The energy clause depends on a stipulated 0.02 J per prevented loss; the break-even is reported so the verdict can be recomputed for any other value. Thresholds were frozen from a pilot seed, and the controller gains ($\zeta = 1$, $\omega = 3$) were fixed before the evaluation but not optimised; a faster rotation would reduce the CDR gap and increase the peak churn, and the Pareto frontier is not mapped here. Finally, one topology family and one attack volume are tested.

7 Conclusion

We replaced the discrete dimension jump of DDMR with a continuous rotation of the metric space and measured the exchange rate. Under a sustained DDoS, CRMR keeps legitimate delivery within 2.4 points of DDMR and 23.4 points above static T-NDN while removing 84% of the peak reconfiguration, with no latency penalty and a passing energy accounting. Under flapping bursts the rotation cannot complete inside the burst and costs 11.2 points; the burst-duration sweep locates the boundary and shows that the worst case is a burst matching the control loop's settling time, not the shortest one. Persistent homology separates the attack as a component, not as a hole. The right reading is not that continuity is free: it is that continuity is a tunable trade between reconfiguration smoothness and delivery under attacks shorter than the rotation, and the trade can be measured, priced and designed.

Reproducibility. The simulator, the frozen pre-registration, the evaluation seeds, the per-seed CSVs and the figure-generation code are in the accompanying repository: `simulation/` (the CRMR model and the scenario), `tools/` (the benchmark) and `docs/crmr-predictions.md`. Every number in this paper is reproduced by that benchmark.

A Pre-registration and pilot

The evaluation follows the frozen protocol in `docs/crmr-predictions.md`. A pilot on seed 2026 was used to fix the guardrails after an initial draft proposed a 2-point sustained gap; the pilot showed a settling-time cost of about three points, and the frozen guardrail became 5 points. The pilot seed is excluded from the 30 evaluation seeds (2030–2059). The flapping gap was declared a measured boundary in advance, not a pass criterion, precisely because the rotation's settling time and the burst length interact; the sweep in Fig. 4 was pre-registered as the way to locate that interaction. No parameter, threshold or seed was changed after the evaluation run.

References

[1] L. Zhang, A. Afanasyev, J. Burke, V. Jacobson, k. claffy, P. Crowley, C. Papadopoulos, L. Wang, and B. Zhang, "Named Data Networking," *ACM SIGCOMM Computer Communication Review*, vol. 44, no. 3, pp. 66–73, 2014. doi:10.1145/2656877.2656887.

[2] A. K. M. M. Hoque, S. O. Amin, A. Alyyan, B. Zhang, L. Zhang, and L. Wang, "NLSR: Named-data Link State Routing Protocol," in *Proc. 3rd ACM SIGCOMM Workshop on Information-Centric Networking (ICN)*, 2013. doi:10.1145/2491224.2491231.

[3] G. Pei, M. Gerla, and T.-W. Chen, "Fisheye State Routing: A Routing Scheme for Ad Hoc Wireless Networks," in *Proc. IEEE International Conference on Communications (ICC)*, 2000. doi:10.1109/ICC.2000.853066.

[4] D. Milić and T. Braun, "Fisheye: Topology Aware Choice of Peers for Overlay Networks," in *Proc. IEEE 34th Conference on Local Computer Networks (LCN)*, 2009. doi:10.1109/LCN.2009.5355168.

[5] H. Edelsbrunner, D. Letscher, and A. Zomorodian, "Topological Persistence and Simplification," *Discrete & Computational Geometry*, vol. 28, no. 4, pp. 511–533, 2002. doi:10.1007/s00454-002-2885-2.

[6] M. Nikolic, *7G Backhaul Simulation: T-NDN and DDMR technical reports*, 2026. Accompanying repository, `docs/tndn.md`, `docs/ddmr.md`, `docs/crmr-predictions.md`.